Dr. RVR
INSTITUTE OF TECHNOLOGY
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# DR.RVR NRI INSTITUTE OF TECHNOLOGY DEEMED TO BE UNIVERSITY

Department of Computer Science and Engineering

## ACTIVITY-BASED LEARNING REPORT

## WEB DEVELOPMENT




Submitted in partial fulfillment of the Web Development Activity-Based Learning assignment

| | |
|---|---|
| **Student Name:** | ***Suma Sri Pasupuleti*** |
| **Roll Number:** | ***26000050*** |
| **Branch / Section:** | ***CSE 1*** |
| **Academic Year:** | ***2026–2027*** |
| **Faculty Name:** | ***Krishnah Sir*** |

## DECLARATION

**I am Suma Sri Pasupuleti, a student of {CSE - 1}, hereby declare that the Web Development Activity-Based Learning assignment submitted by me has been completed as part of my academic learning.**

**This report documents the three selected case studies, their objectives, design, implementation, screenshots, testing, and results. The work presented reflects my understanding and practical application of web development concepts.**

**I have made efforts to ensure that the information presented in this report is accurate and that the project work and external resources used are appropriately acknowledged.**

**Student Name: Suma Sri Pasupuleti**

**Roll Number: 26000050**

**Signature: P. Suma Sri**

**Date: 26-10-26**

# ACKNOWLEDGMENT

**I would like to express my sincere gratitude to the management and faculty of Dr. RVR NRI Institute of Technology Deemed To Be University for providing me with the opportunity to complete this Web Development Activity-Based Learning assignment.**

**I am thankful to my Web Development faculty for the guidance, encouragement, and support provided throughout the completion of this assignment.**

**This activity helped me understand the practical application of HTML and CSS and, where applicable, JavaScript. It also gave me an opportunity to improve my webpage design, coding, debugging, testing, and project organization skills.**

**Finally, I would like to thank everyone who supported me in completing this assignment successfully.**

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# INTRODUCTION

Web development is the process of creating websites and web applications that can be accessed through a web browser. It involves designing webpage layouts, organizing content, applying styles, and adding interactive features when required.

HTML (HyperText Markup Language) is used to structure webpage content. CSS (Cascading Style Sheets) is used to control the appearance, layout, colours, spacing, and responsiveness of webpages. JavaScript can be used to implement interactive features such as form validation, menus, dynamic content, and button actions.

Activity-Based Learning provides an opportunity to apply theoretical concepts to practical problems. Through this assignment, three selected web development case studies are designed, implemented, tested, and documented.

**2. OBJECTIVES**

The main objectives of this assignment are:

1. To understand the basic principles of web development.
2. To create structured webpages using HTML.
3. To design visually appealing layouts using CSS.
4. To develop responsive webpages that adapt to different screen sizes.
5. To implement interactive features using JavaScript where required.
6. To test webpages and correct implementation errors.
7. To organize project files in a structured manner.
8. To upload the completed case studies to GitHub.
9. To improve practical problem-solving and web development skills.

## 3. SCOPE OF THE ASSIGNMENT

This assignment covers the development of three webpages or websites based on the case studies selected from the list provided by the faculty.

Each case study includes a problem statement, objectives, design approach, implementation details, screenshots, testing, and results. The completed projects are organized in a GitHub repository along with a README file and a consolidated PDF repor

## 4. TECHNOLOGIES AND TOOLS

**HTML5**: Used to create the structure and content of webpages.

**CSS3**: Used to style webpages, arrange elements, apply colours, create layouts, and support responsive design.

**Web Browser**: Used to open, run, and test the webpages.

**Code Editor**: Used to create and edit the project files.

**GitHub**: Used to store and organize the completed project files and README.

**CASE NUMBER – 1**

**CASE NAME:** PERSONAL PORTFOLIO WEBSITE

**EXPLANATION:** A personal portfolio is a website that presents an individual's profile, educational background, technical skills, projects, interests, achievements, and contact information in one place.

The purpose of this case study is to design and develop a personal portfolio website that organizes personal and academic information in a clear, attractive, and accessible format. The website should provide easy navigation between sections and a responsive layout for different screen sizes.

**OBJECTIVE**

Include these points:

- To create a personal website using HTML and CSS.
- To display personal information and educational qualifications.
- To present technical skills and project details.
- To provide navigation links to different sections.
- To include resume and contact options.
- To apply styling, hover effects, and CSS animations.
- To make the webpage responsive for smaller screens.

**Software and technologies used**

| Tool / technology | Purpose |
|---|---|
| HTML5 | Structure of the webpage |
| CSS3 | Styling, layout and animations |
| Visual Studio Code or your actual editor | Writing and editing code |
| Google Chrome or your actual browser | Running and testing the website |
| GitHub | Hosting project files in a repository |

**DESIGN AND IMPLEMENTATION**

The personal portfolio website is divided into multiple sections to make the information easy to understand and navigate.

**1. Navigation Bar:** Contains links to Home, About, Education, Skills, Projects, Interests, Resume, and Contact sections. The links use HTML anchor references to navigate within the same webpage.

**2. Home Section:** Introduces the student, displays the portfolio heading and career interest, and provides a button to explore additional information.

**3. About Section:** Provides a short personal introduction and describes the student's interest in technology, programming, and web development.

**4. Education Section:** Displays the college, intermediate, and school education details in separate cards.

**5. Skills Section:** Presents the technical skills included in the portfolio, such as HTML, CSS, and Python.

**6. Projects Section:** Displays project cards with titles and descriptions.

**7. Interests and Achievements Section:** Provides information about hobbies, interests, certificates, and achievements where applicable.

**8. Resume Section:** Contains a link to download the resume PDF. The link works when the correct resume file is available at the specified path.

**9. Contact Section:** Displays the student's email address, phone number, and location, along with an email link.

**10. Footer:** Displays the copyright notice and portfolio identification.

CSS is used to apply a dark background, purple accent colors, cards, spacing, hover effects, animations, and a responsive layout. The media query adjusts the layout for smaller screens.

**KEY FEATURES:**

**-** Single-page portfolio with section navigation.

- Dark background and purple accent theme.

- Card-based education, skills, and project sections.

- Hover effects on navigation links and cards.

- Animated home content and profile illustration.

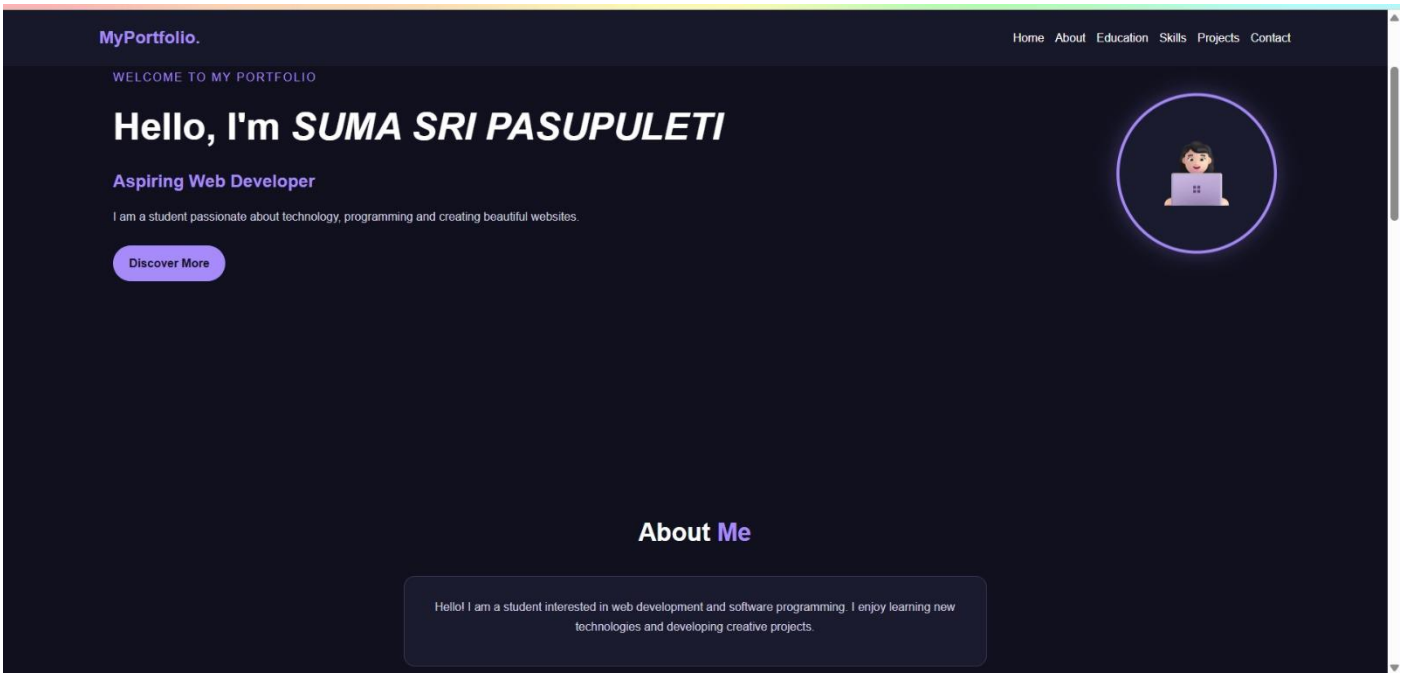
- Resume download link.

- Email contact link.

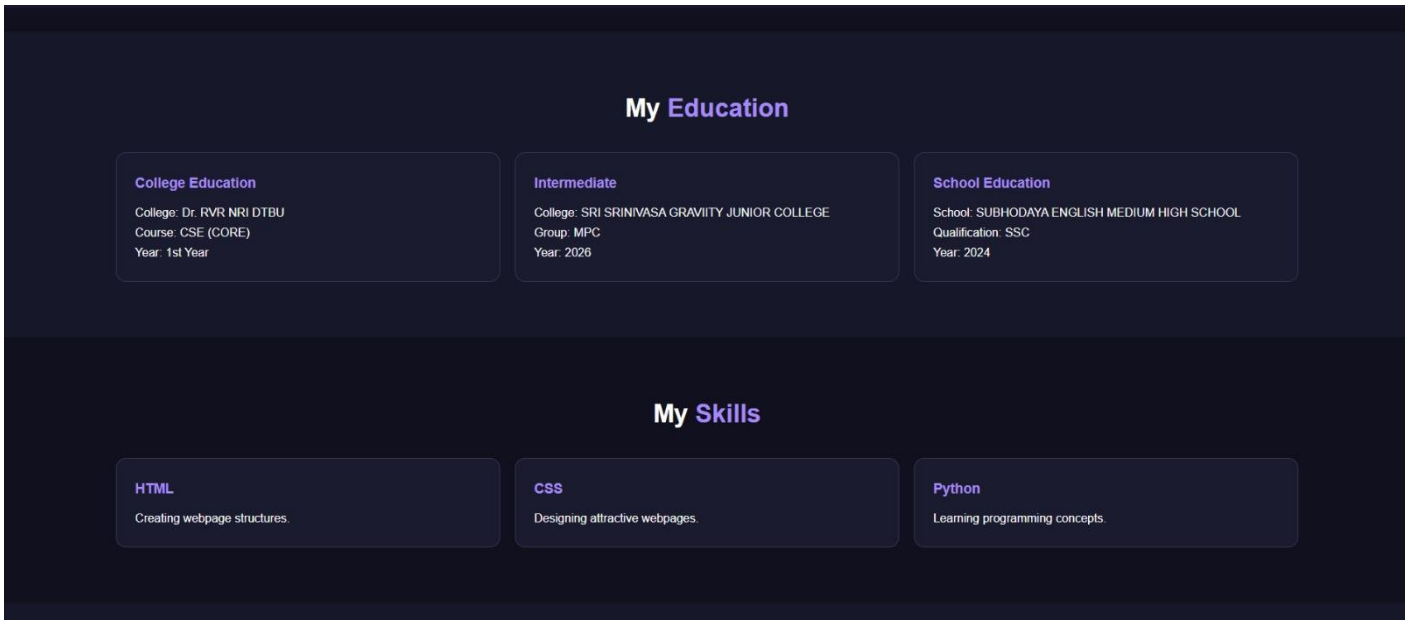
- Responsive layout for smaller screens.

## SCREEN SHOTS OF THE WEBSITE:

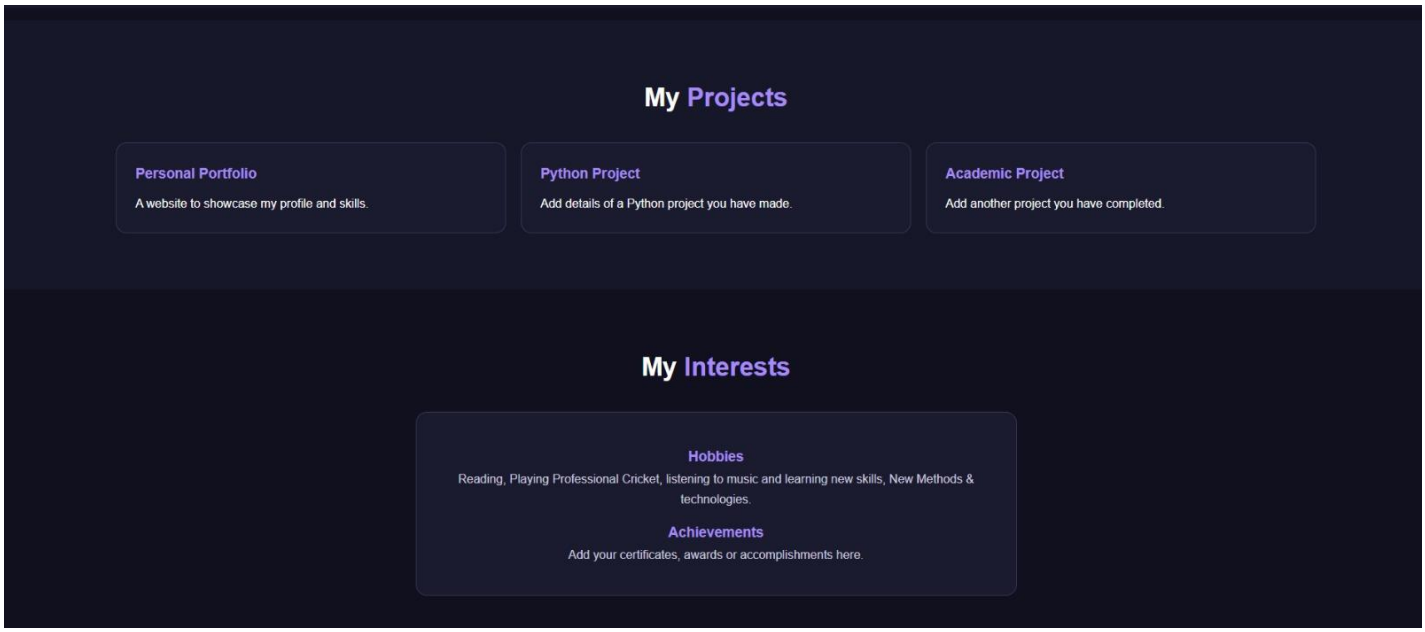
**SCREEN SHOT – 1 (Homepage showing the navigation bar, welcome message, introduction and profile illustration.)**



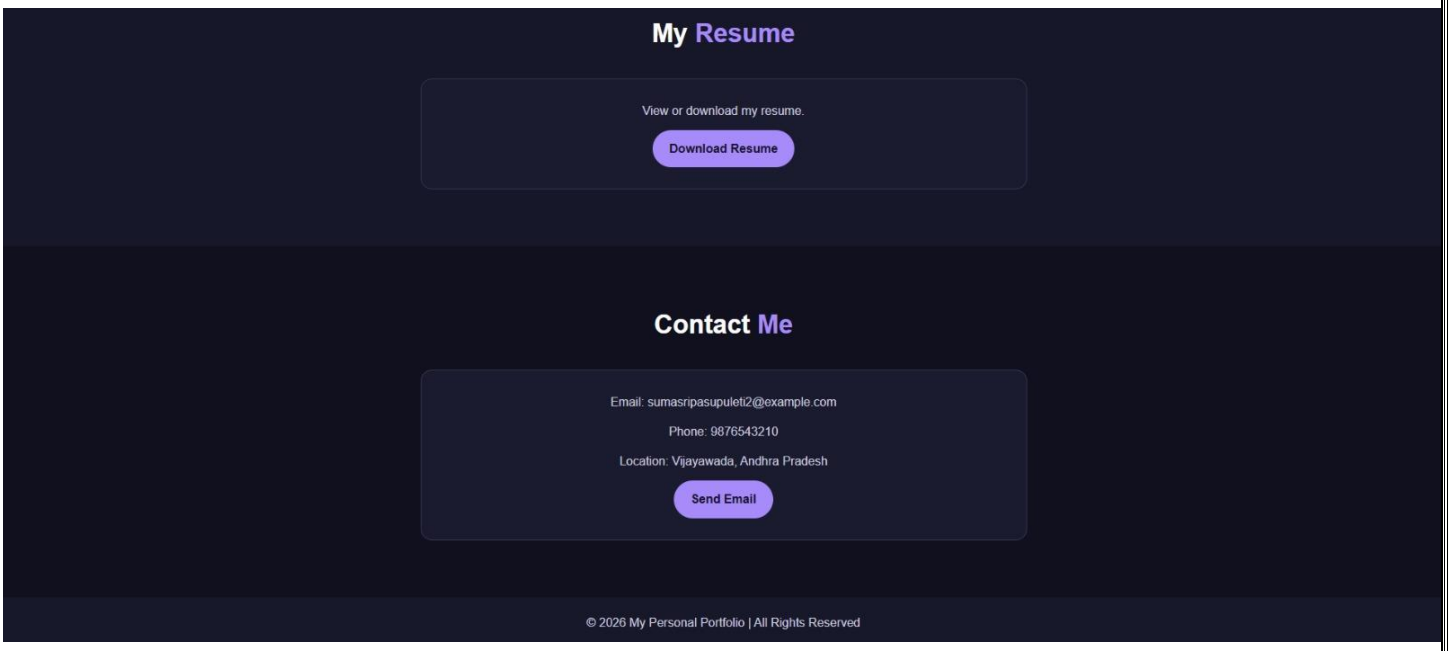

**SCREEN SHOT – 2 (Education details and technical skills presented using cards.)**



**SCREEN SHOT – 3 (Project information, hobbies and achievements.)**




**SCREEN SHOT – 4 (Resume download option and contact information.)**



**LEARNING OUT COMES:**

**-** Understood the basic HTML document structure.

- Learned how to create navigation using anchor links.

- Applied CSS selectors, variables, colours, spacing, and borders.

- Used Flexbox and CSS Grid for page layouts.

- Applied transitions, hover effects, and animations.

- Practiced responsive design using media queries.

- Learned to test a website in a browser and correct errors.

**RESULT:**

The personal portfolio website was developed using HTML5 and CSS3. It organizes the student's profile, educational background, skills, projects, interests, resume link, and contact information into clearly separated sections.

CSS styling, card layouts, hover effects, animations, and responsive design techniques were applied to improve the presentation and usability of the website. The completed webpage was opened in a browser and tested for navigation and layout behaviour.

The project provided practical experience in structuring webpages, applying styles, organizing content, and testing a website.

**CASE NUMBER – 3**

**CASE NAME:** ONLINE LEARNING PLATFORM INTERFACE

**EXPLANATION:** An online learning platform is a website that provides information about online courses, educational categories, instructors, and learning progress. It helps learners explore different subjects, view course details, and identify learning opportunities through a structured and user-friendly interface.

The purpose of this case study is to design and develop an online learning platform interface named LearnHub using HTML5 and CSS3. The website presents course categories, popular courses, instructor information, learning progress indicators, and calls to action in an organized format. It uses a clean layout, purple accent colors, cards, and responsive design techniques to improve the visual appearance and user experience.

**OBJECTIVES': -**

To create an online learning platform interface using HTML and CSS.

To display educational courses and categories in an organized manner.

To present instructor information and course details.

To design a user-friendly navigation bar.

To display learning progress indicators.

To apply CSS styling, card layouts, colors, and visual effects.

To organize educational information into clearly defined sections.

To develop a visually appealing webpage with responsive design where implemented.

**SOFTWARE AND TECHNOLOGIES USED: -**

| Tool / Technology | Purpose |
|---|---|
| HTML5 | Structure and organization of the webpage |
| CSS3 | Styling, layout, colours, spacing, and visual effects |
| Visual Studio Code or the actual editor used | Writing and editing code |
| Google Chrome or the actual browser used | Running and testing the website |
| GitHub (if used) | Hosting project files in a repository |

**DESIGN AND IMPLEMENTATION:**

The Learn Hub online learning platform is divided into multiple sections to make educational information easy to understand and navigate.

**1. Navigation Bar:** Contains the Learn Hub logo and links to Home, Categories, Courses, Instructors, and Contact. An “Explore Courses” button is also provided to direct users toward the course section.

**2. Home Section:** Introduces the platform with the heading “Upgrade Your Skills With Online Learning.” It includes a short description, “Explore Courses” and “Learn More” buttons, and statistics showing 50+ courses, 20+ instructors, and 1000+ learners.

**3. Course Categories Section:** Displays categories such as Web Development, Programming, UI/UX Design, Aptitude Development, and Artificial Intelligence. Each category is presented in a separate card with an icon and the number of available courses.

**4. Popular Courses Section:** Displays course cards for HTML & CSS Fundamentals, Python Programming, UI/UX Web Design, and Aptitude Development. Each card includes a difficulty level, course icon, number of lessons, rating, course title, description, instructor information, price, and an “Enroll Now” button.

**5. Learning Progress Section:** Displays the heading “Your Learning Progress” and uses progress bars to represent learning completion percentages for Aptitude Development, HTML & CSS Fundamentals, Python Programming, and UI/UX Web Design. The screenshot indicates that these progress values are sample indicators for demonstration purposes.

**6. Instructors Section:** Presents instructor information in separate cards. The displayed instructors include Krishnan Sir, Venu Gopal Sir, Indraja Mam, and another instructor card for an aptitude trainer. Their teaching areas and rating indicators are also displayed.

**7. Call-to-Action Section:** Displays the message “Ready to Learn Something New?” with a short description and a “Browse Courses” button to encourage learners to explore available courses.

**8. Footer Section:** Displays the Learn Hub logo, a short tagline, navigation links, and a copyright notice identifying the project.

**9. Course and Learning Information:** Organizes course descriptions, lesson counts, difficulty levels, ratings, instructor details, and sample pricing into readable cards for convenient viewing.

**10. CSS Styling and Layout:** CSS is used to create a light background, purple accent colors, gradient sections, rounded cards, spacing, typography, and button styles. Grid-based layouts and responsive styling techniques are used wherever implemented to organize content and support different screen sizes.

The website is implemented using HTML5 and CSS3, with HTML providing the structure and CSS controlling the visual appearance and layout.

**KEY FEATURES: -**

LearnHub branding and navigation bar.

Attractive homepage with an introduction to online learning.

Course statistics showing courses, instructors, and learners.

Five educational course categories.

Popular course cards with lesson counts and ratings.

Course difficulty labels such as Beginner and Intermediate.

Instructor information and teaching specializations.

Learning progress bars with sample completion percentages.

Course pricing and “Enroll Now” buttons.

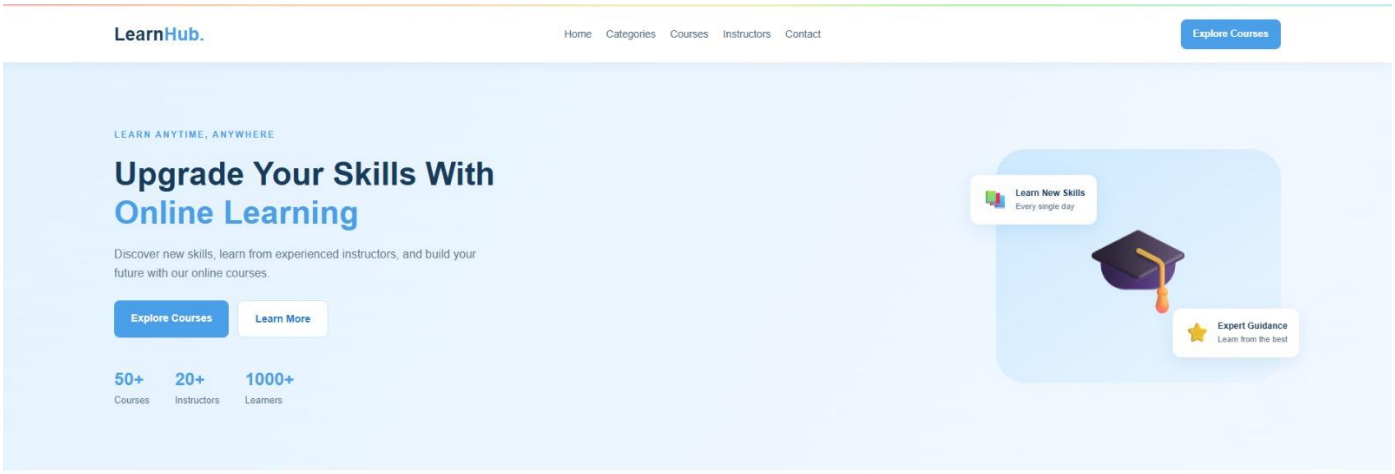
Call-to-action section with a “Browse Courses” button.

Consistent purple-themed design with rounded cards and gradient backgrounds.

Organized footer with navigation links and copyright information.

**SCREEN SHOTS OF THE WEBSITE:**

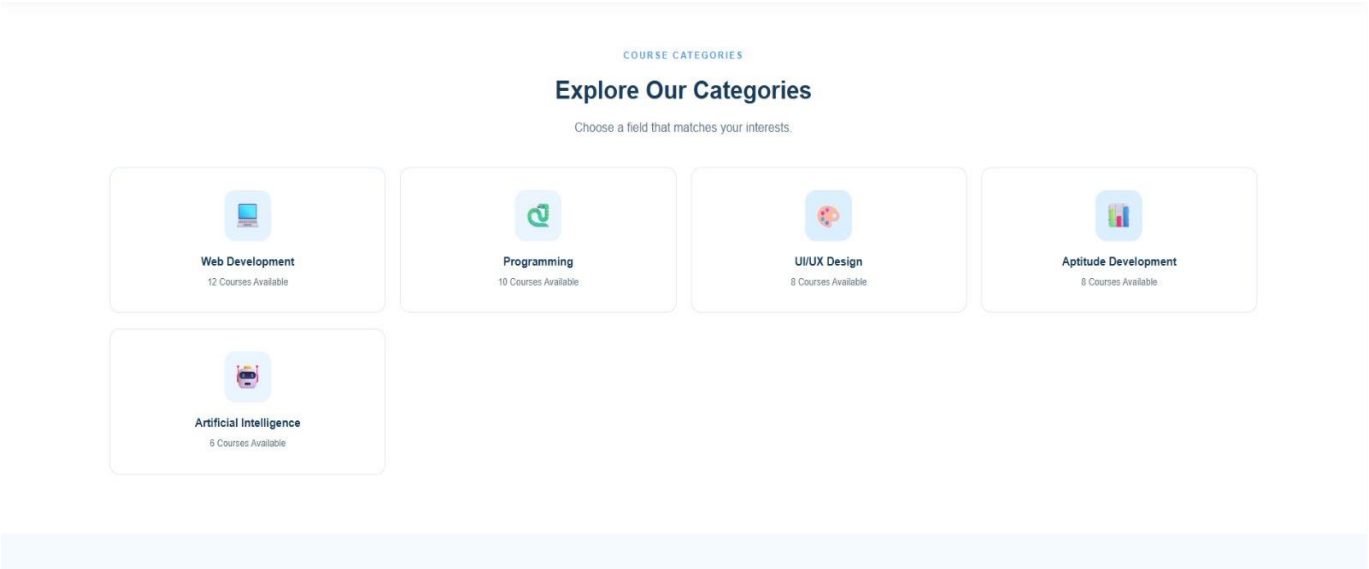
**SCREEN SHOT – 1 (Homepage displaying the LearnHub navigation bar, welcome message, introductory content, course statistics, and educational categories.)**



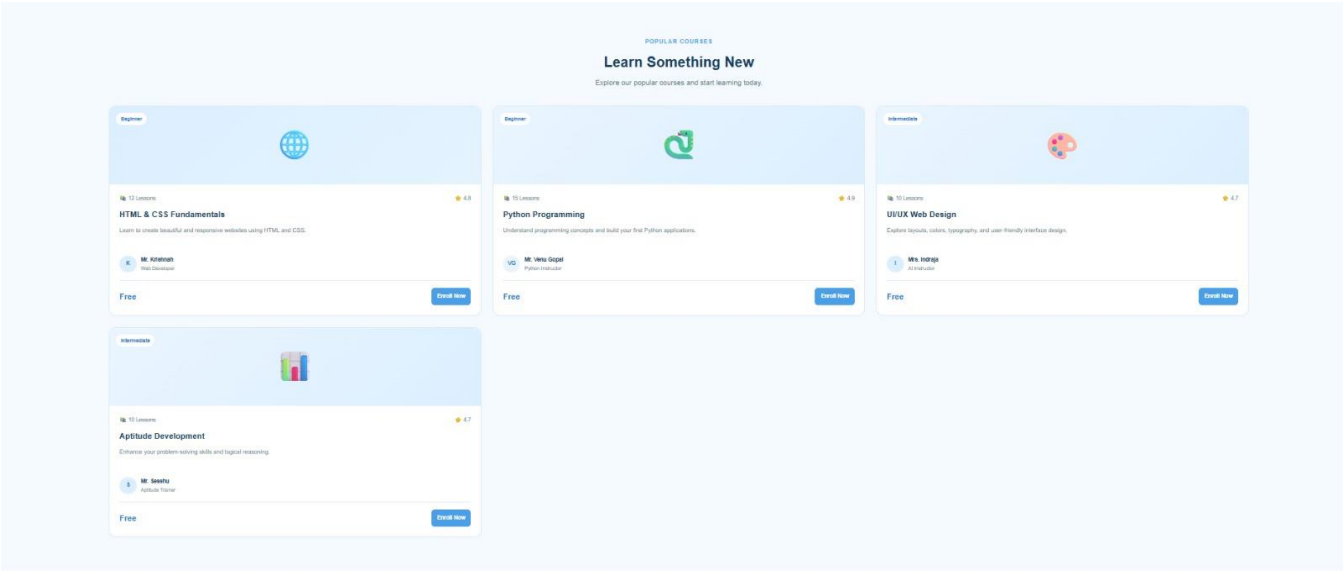




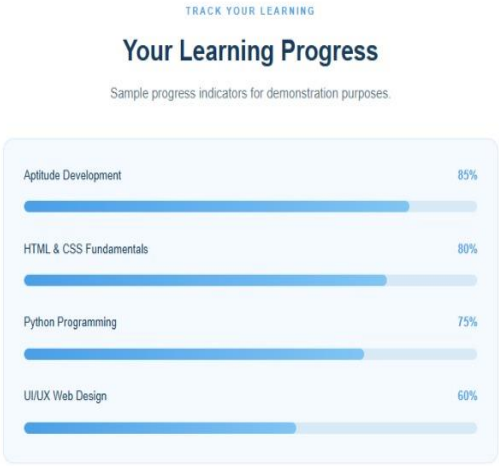
**SCREEN SHOT – 2 (Popular courses displayed using cards containing course titles, difficulty levels, lesson counts, ratings, instructor details, prices, and enrolment buttons.)**



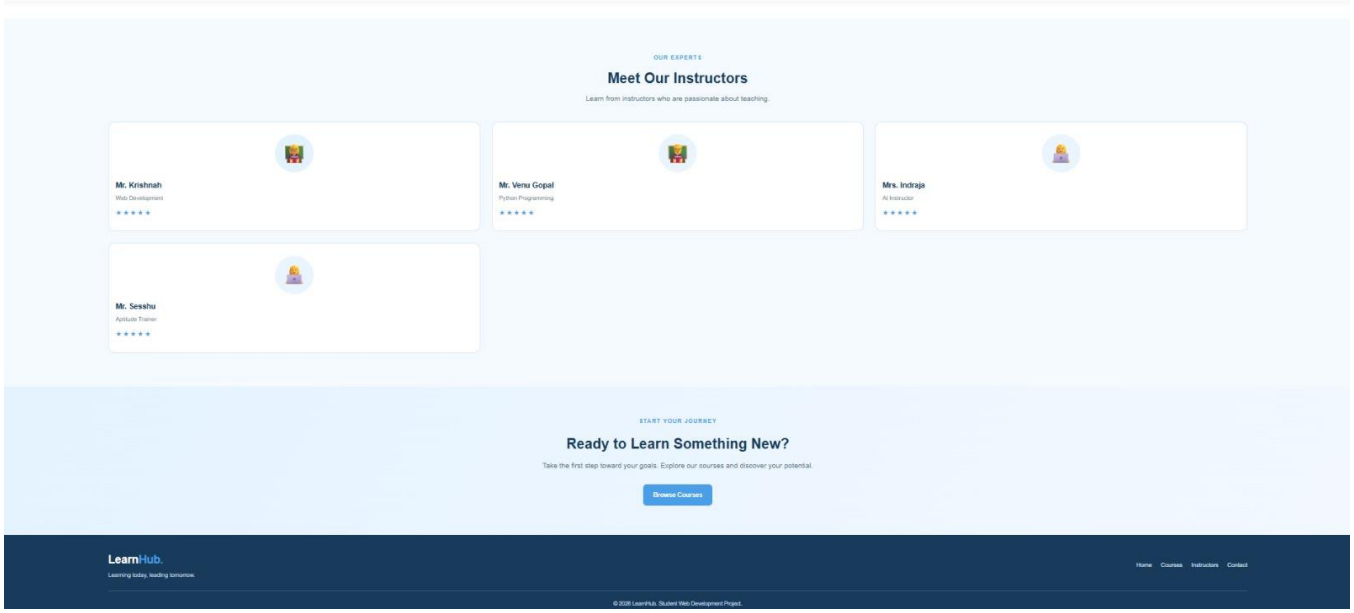
**SCREEN SHOT – 3 (Course categories including Web Development, Programming, UI/UX Design, Aptitude Development, and Artificial Intelligence, arranged in separate cards.)**




**SCREEN SHOT – 4 ( The learning progress section with sample progress indicators.)**



**SCREEN SHOT – 5 ( Instructor cards displaying names and teaching areas, followed by the call-to-action section and website footer.)**




**LEARNING OUT COMES: -**

Understood the basic structure of an HTML document.

Learned to design a structured educational website interface.

Practiced creating navigation links and section layouts.

Applied CSS selectors, colors, typography, margins, padding, and borders.

Learned to arrange course information using cards and grid layouts.

Understood how to present course ratings, lesson counts, and instructor details.

Practiced designing progress bars and course information panels.

Learned to improve visual appearance through gradients, spacing, and rounded corners.

Gained experience in organizing large amounts of information into separate sections.

Improved understanding of user interface design and front-end web development.

**RESULT: -**

The Online Learning Platform Interface (LearnHub) was designed using HTML5 and CSS3. The website presents educational categories, course information, instructor details, sample learning progress, and navigation options in a structured and visually appealing format.

CSS styling, card-based layouts, progress bars, gradient backgrounds, and consistent typography were used to improve the appearance and usability of the webpage. The project provided practical experience in webpage structuring, front-end design, content organization, and browser-based testing.

The completed interface demonstrates the fundamental principles of designing an educational website and provides a foundation for future development into a functional online learning platform.

**CASE NUMBER – 11**

**CASE NAME: HOSPITAL AND HEALTHCARE SERVICES WEB SITE**

**EXPLANATION:** A Hospital and Healthcare Services Website is a website that presents information about a hospital, its medical departments, doctors, specialties, facilities, visiting hours, emergency contact details, and appointment request options.

The purpose of this case study is to design and develop a hospital website that organizes healthcare information in a clear, attractive, and accessible format. The website provides easy navigation between sections and uses a responsive layout to support different screen sizes.

**OBJECTIVES: -**

To create a hospital website using HTML and CSS.

To display hospital information, departments, and medical specialties.

To present doctor profiles and healthcare facilities.

To provide visiting hours and emergency contact details.

To create an accessible appointment request form.

To apply attractive styling, hover effects, and CSS transitions.

To make the webpage responsive for smaller screens.

To improve the readability and usability of healthcare information.

**SOFT WARE AND TECHNOLOGIES USED**

| Tool / Technology | Purpose |
|---|---|
| HTML5 | Structure of the webpage |
| CSS3 | Styling, layout, colours, and animations |
| Visual Studio Code | Writing and editing code |
| Google Chrome | Running and testing the website |
| GitHub | Hosting project files in a repositor |

**DESIGN AND IMPLEMENTATION: -**

The Hospital and Healthcare Services Website is divided into multiple sections to make healthcare information easy to understand and navigate.

**1. Navigation Bar:** Contains links to Home, About, Departments, Doctors, Facilities, and Contact sections. The links use HTML anchor references to navigate within the same webpage.

**2. Home Section:** Introduces the hospital and displays a welcome message, healthcare information, a hospital image, and buttons to explore services or request an appointment.

**3. About Section:** Provides information about the hospital's healthcare approach, patient-focused services, and commitment to providing quality care.

**4. Departments and Specialties Section:** Displays medical departments such as Cardiology, General Medicine, Neurology, Paediatrics, Orthopaedics, and Dermatology in separate cards.

**5. Doctors Section:** Presents illustrative doctor profiles with names, medical specialties, photographs, and appointment request links.

**6. Facilities Section:** Describes hospital facilities and services, including emergency services, diagnostic support, inpatient care, and patient accessibility.

**7. Visiting Hours and Contact Section:** Displays sample visiting hours, emergency contact information, email address, and hospital location.

**8. Appointment Request Section:** Contains a form with fields for the patient's name, phone number, email address, department, preferred doctor, appointment date, preferred time, and an additional message. HTML form validation and accessible labels are used to improve usability.

**9. Call-to-Action Section:** Encourages visitors to request an appointment through a clearly visible button.

**10. Footer:** Displays quick navigation links, patient services, contact information, location, and copyright information.

CSS is used to apply a navy blue, teal, and white colour combination, card-based layouts, rounded corners, shadows, spacing, hover effects, and responsive styling. CSS media queries adjust the webpage layout for tablets and smaller screens.

**KEY FEATURES: -**

Single-page hospital website with section navigation.

Professional colour theme: A clean combination of *dark navy blue, teal, light blue-grey, and white*, with soft aqua accents and subtle borders, creating a modern, calm, and healthcare-friendly appearance.Hospital introduction and medical service information.

Department and specialty cards.

Doctor profiles with photographs.

Hospital facilities and visiting hours.

Emergency contact information.

Appointment request form with accessible labels.

Form validation using HTML attributes.

Hover effects and smooth transitions.

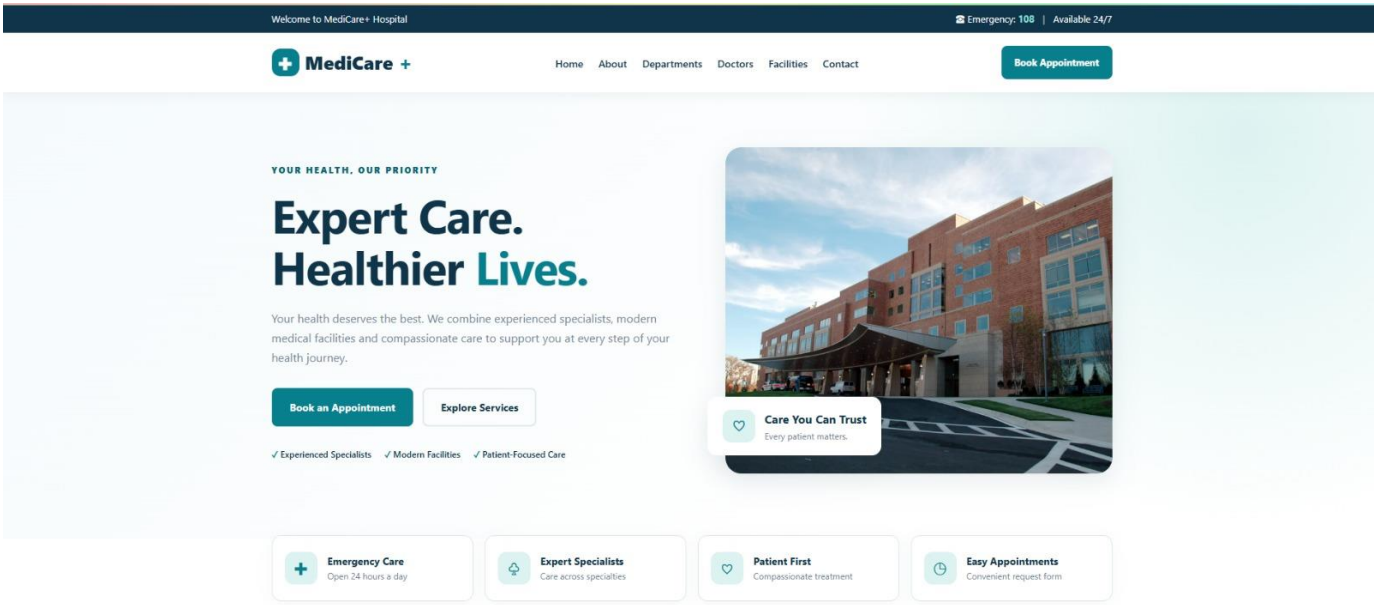
Responsive layout for smaller screens.

Semantic HTML elements and alternative text for images.

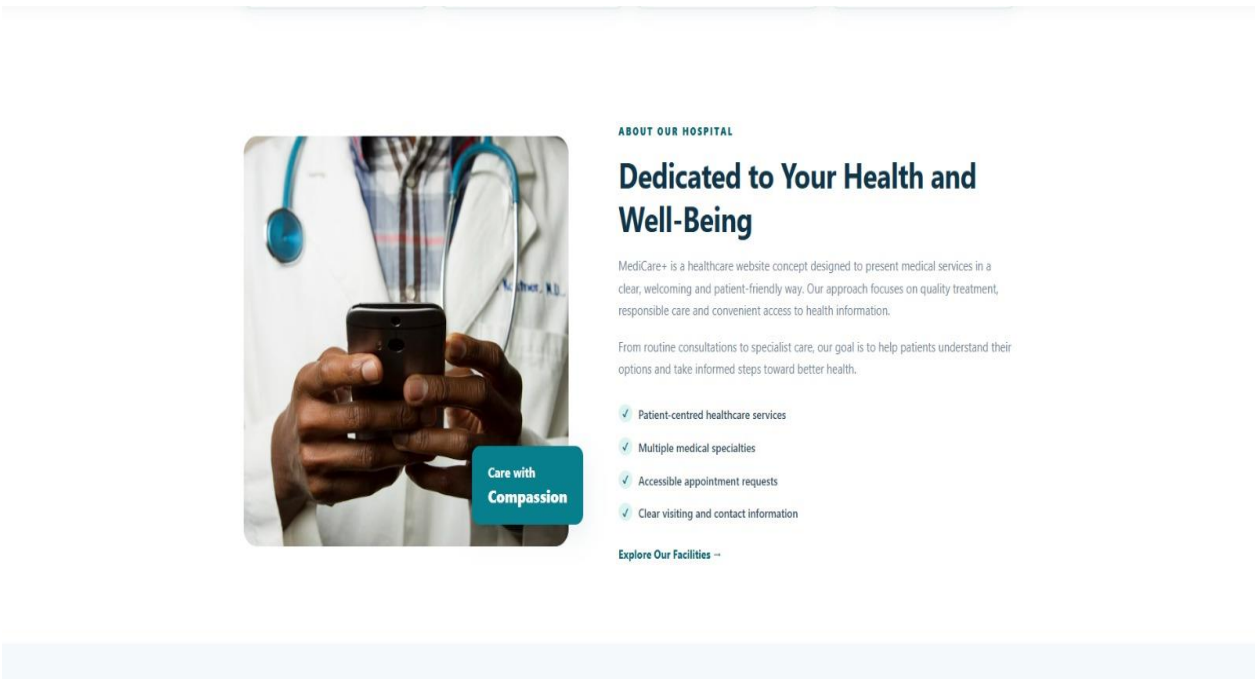
Email and telephone links for convenient contact.

**SCREEN SHOTS OF WEB SITE**

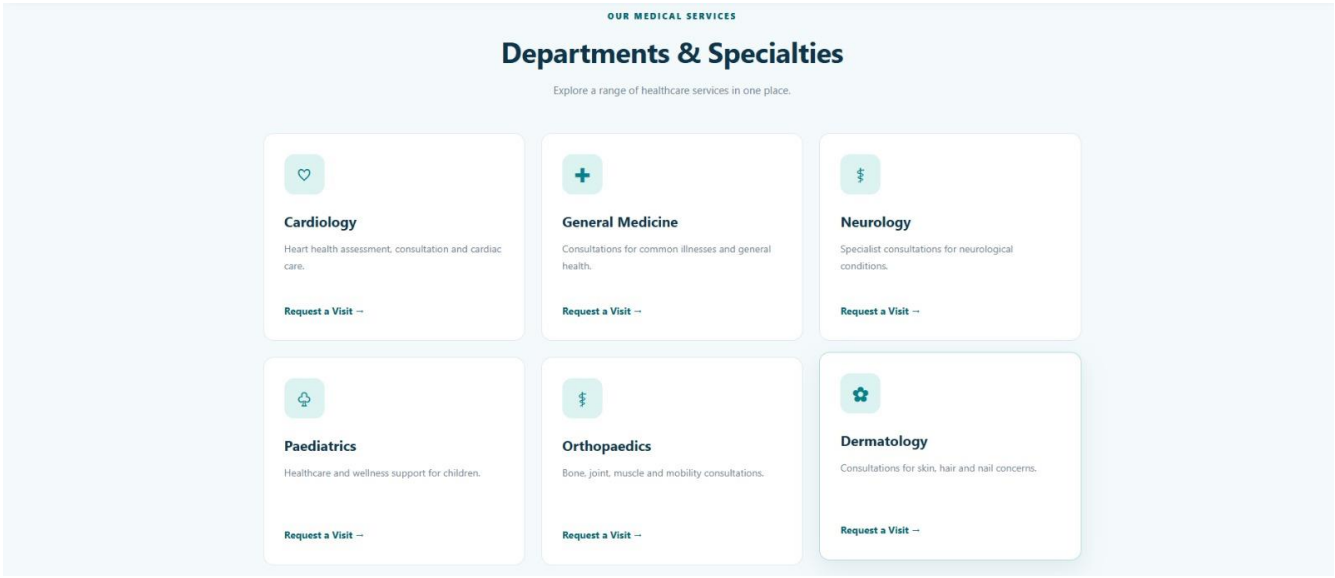
**SCREEN SHOT – 1 ( Homepage showing the navigation bar, hospital welcome message, introduction, hospital image, and appointment button )**



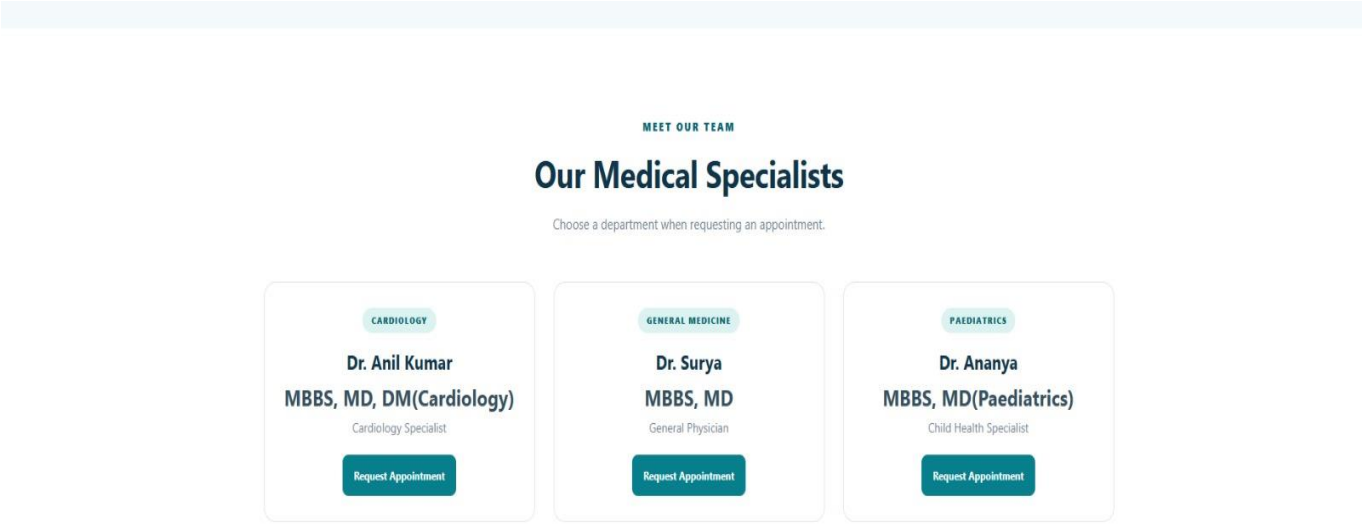

**SCREEN SHOT – 2 ( Quick information cards and About Hospital section presenting the hospital's services, healthcare approach, and patient-focused care.)**



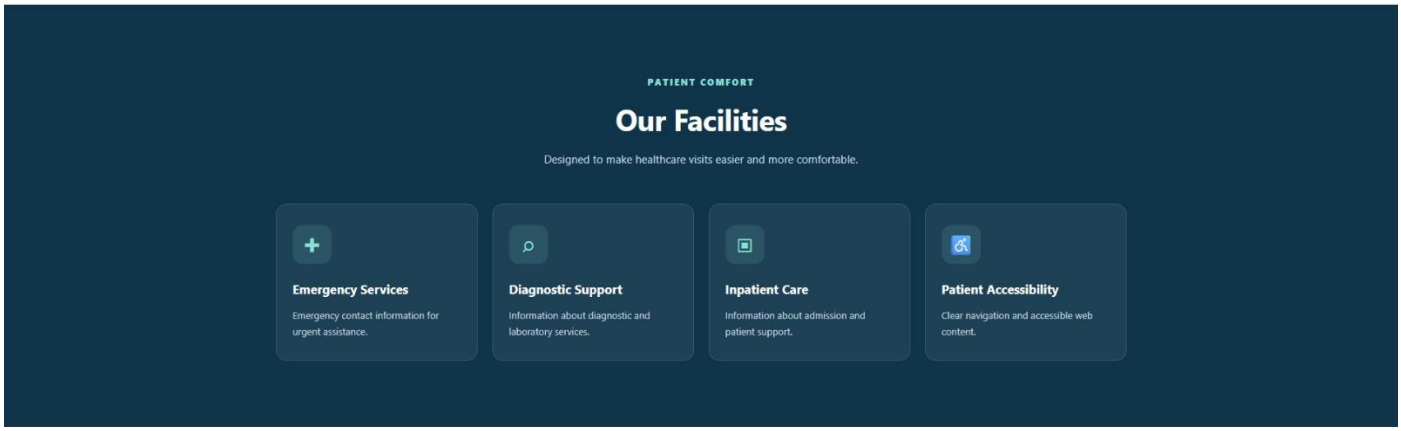
**SCREEN SHOT – 3 ( Departments and Specialties section displaying medical services such as Cardiology, General Medicine, Neurology, Paediatrics, Orthopaedics, and Dermatology.)**



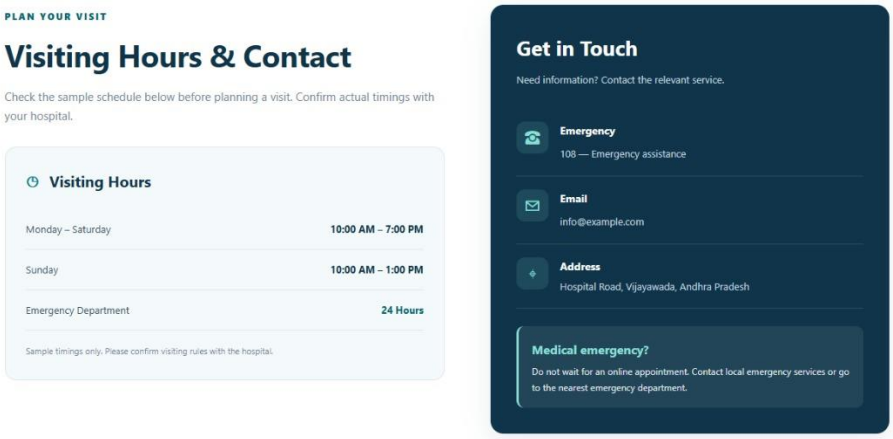

**SCREEN SHOT – 4 ( Doctors section showing specialist profiles, doctor names, medical specialties, and appointment request buttons.)**



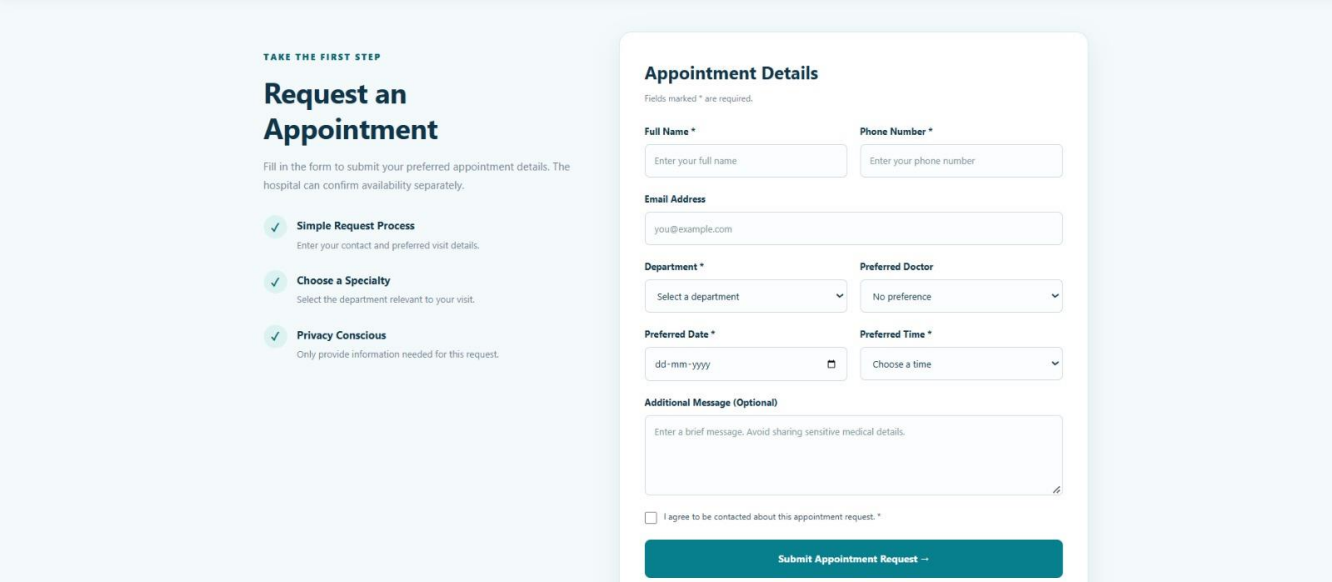
**SCREEN SHOT – 5 ( Facilities section presenting emergency services, diagnostic support, inpatient care, and patient accessibility features.)**



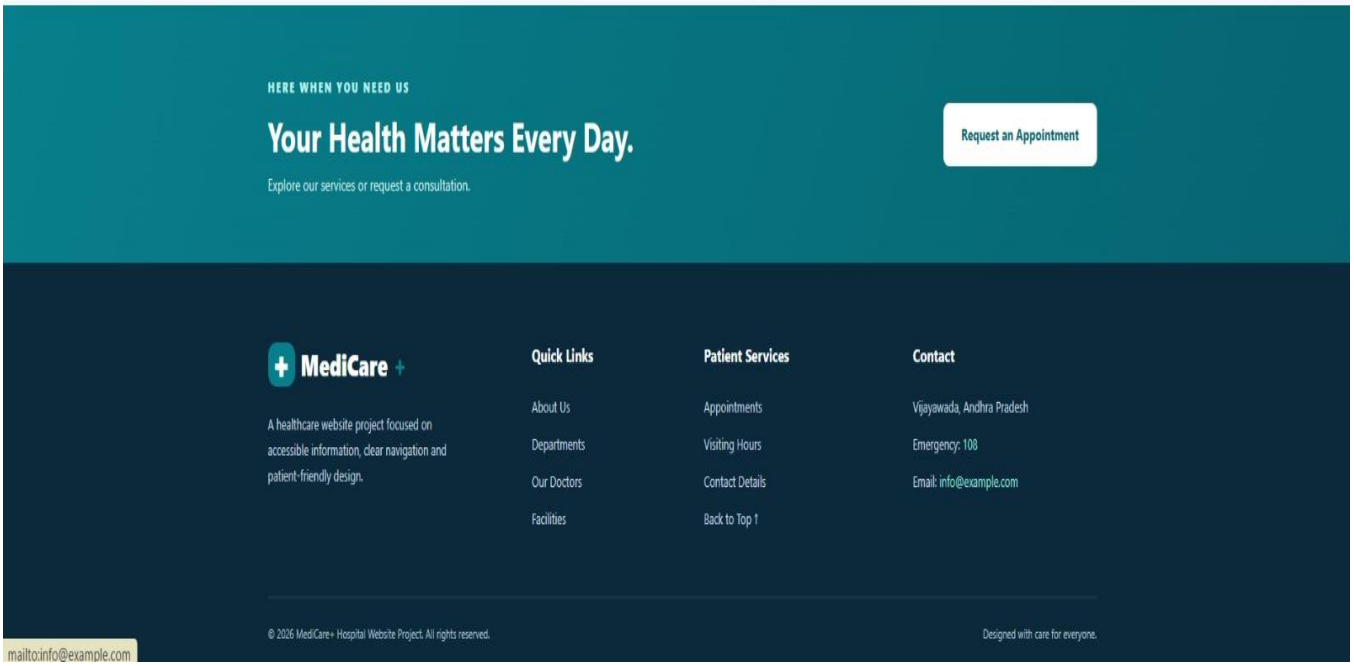

**SCREEN SHOT – 6 ( Visiting Hours and Contact section displaying visiting schedules, emergency contact information, email address, and hospital location.)**



**SCREEN SHOT – 7 ( Appointment Request section showing the patient information fields, department selection, preferred doctor, date and time fields, message box, and consent checkbox)**




**SCREEN SHOT – 8 (Lower appointment form and call-to-action section, followed by the website footer containing quick links, patient services, contact details, and copyright information.)**



**LEARNING OUT COMES**

Understood the basic structure of an HTML document.

Learned to create navigation links using HTML anchor elements.

Practiced semantic HTML elements for organizing website sections

Applied CSS selectors, variables, colours, spacing, borders, and shadows.

Used Flexbox and CSS Grid to arrange content and cards.

Created accessible form controls using labels and required fields.

Applied hover effects and transitions to improve user interaction.

Practiced responsive web design using CSS media queries.

Learned to use images and alternative text appropriately.

Tested webpage navigation, form validation, and layouts in a browser

**RESULT**

The Hospital and Healthcare Services Website was developed using HTML5 and CSS3. It organizes hospital information, medical departments, doctor profiles, healthcare facilities, visiting hours, emergency contact details, and appointment request options into clearly separated sections.

CSS styling, card-based layouts, hover effects, transitions, and responsive design techniques were applied to improve the presentation and usability of the website. Accessible form labels, required fields, descriptive image text, and keyboard focus styles were included to support a more user-friendly experience.

The completed webpage was opened in a browser and tested for navigation, form validation, and responsive layout behaviour. The project provided practical experience in developing structured webpages, designing healthcare interfaces, implementing accessible forms, and testing a responsive website.